

A/B Test Report: Website Traffic Data

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Objective The objective is to determine whether website Variant B changes conversion performance compared with Variant A.

Dataset The dataset contains 2,400 website visits split evenly between Group A and Group B. Each row includes visitor ID, visit date, experiment group, device, traffic channel, session activity, conversion flag, and revenue.

Hypotheses - H0: Conversion rate for Group B equals conversion rate for Group A. - H1: Conversion rate for Group B differs from conversion rate for Group A. - Alpha: 0.05

Group Comparison	Group	Visitors	Conversions	Conversion Rate	Revenue
A	1200	148	12.33%	\$8522.00	
B	1200	186	15.50%	\$11560.11	

Statistical Test A two-proportion z-test was used because the target metric is binary conversion and the experiment compares two independent groups.

- Z-statistic: 2.2410
- P-value: 0.025023
- Absolute lift: 3.17%
- Relative lift: 25.68%
- 95% confidence interval for B
- A: [0.40%, 5.93%]

Decision Reject H0 at alpha = 0.05.

Business Interpretation Variant B shows a higher observed conversion rate than Variant A. Because the confidence interval is above zero and the

p-value is below 0.05, the result supports Variant B as the better-performing experience for conversion.

Interview Questions **What is A/B testing? A/B testing is a controlled experiment that compares two versions of a product, page, or process to determine whether a change improves a selected metric.**

What is a confidence interval? A confidence interval is a range of plausible values for an estimated effect. Here, it estimates the likely conversion-rate difference between Variant B and Variant A.